

Mahtab Faraji

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EDUCATION

- University of Illinois Chicago (UIC)** Jan 2023 - Present
Department of Biomedical Engineering, Chicago, IL, USA
PhD Student in Biomedical Engineering
- University of Science and Technology (UST)** Sep 2015 – Sep 2018
Department of Electrical Engineering
M.Sc. degree in Biomedical Engineering
- Sahand University of Technology (SUT)** Jan 2011 – Aug 2014
Department of Electrical Engineering
B.Sc. degree in Biomedical Engineering

SKILLS SUMMARY

- Programming Languages:** Python (Advanced), MATLAB (Advanced), JavaScript (Intermediate), ReactJS (Intermediate), C# (Familiar), MySQL (Familiar)
- Containerization:** Docker
- Deep Learning (DL) Frameworks:** Pytorch, Tensorflow
- Python Programming Libraries:** OpenCV, Scikit, Numpy, Pandas, Scipy
- Tools:** Git, Google Colab, Conda, Spyder, Pycharm, VScode, Bash
- Operating system:** Linux, Windows

PUBLICATIONS

- Mahtab Faraji**, et al. 2020, Novel approach for automatic mid-diastole frame detection in 2D echocardiography sequences for performing planimetry of the mitral valve orifice, IET Image Processing
Description: This project proposes an automated method for assessing Mitral Valve Stenosis (MS) severity by measuring the mitral valve orifice area using the dimensional reduction and machine learning algorithms.
- Mahtab Faraji**, et al. 2024, Trends, Challenges, and Future Directions in DL for Glaucoma: A Systematic Review, Under review in Translational Vision Science Technology (TVST)
Description: This paper systematically reviewed recent literature in the glaucoma detection using the DL models and provide a direction to the future research.
- Rogério G. Nespolo, George R. Nahass, **Mahtab Faraji**, et al. 2024, Assessing Vitreoretinal Surgical Training Experience by Leveraging Instrument Maneuvers and Visual Attention with DL Neural Networks, ARVO Annual Meeting Abstract
Description: This project develops a DL platform to assess vitreoretinal surgeons' experience by analyzing their instrument maneuvers and gaze tracking during tasks on a surgical simulator.
- Mahtab Faraji**, et al. 2024, Analysis of Out-of- Distribution Factors to Detect Iris and Pupil Using Cataract Surgical Images, CMIMI 2024 Annual Meeting Abstract
Description: This project investigates how Out-Of-Distribution (OOD) data affects DL models for iris and pupil detection in cataract surgery. It focuses on the impact of variations in data acquisition and preprocessing on model performance and generalizability.

EXPERIENCE

- Research and Development- Intern** May 2024 – Aug 2024
Pfizer, San Diego, CA
 - * Optimized large-scale histopathology image processing workflows to improve efficiency and accuracy using advanced image analysis techniques.
 - * Applied Generative Adversarial Networks (GANs) for color normalization of histopathology images to enhance consistency and reliability in downstream analysis.

- * Used AI-based tools, such as QuPath, to detect abnormalities in histopathology images.
- * Contributed to a research paper related to histopathology image color normalization.
- **Research Assistant - Deep Learning/Computer Vision** Jan 2023 – Present
Artificial Intelligence In Ophthalmology Center, Chicago, IL
 - * Participated in a research project aiming at creating a comprehensive review paper that explores the application of DL models in the field of glaucoma detection.
 - * Analyzing Out-of-Distribution factors in DL models trained on the ophthalmic images.
 - * Implementing semi-supervised segmentation techniques on the ophthalmic images.
- **Research Assistant - Deep Learning/Computer Vision** Sep 2018 - Aug 2021
University of Science and Technology
 - * Collected 2-D echocardiography videos from the Cardiovascular-Medical and Research Center for research purposes.
 - * Applied image processing techniques like filtering, segmentation, and object detection to analyze echocardiography images.
 - * Utilized dimensionality reduction methods, including PCA, kernel-PCA, and LLE, to identify the mid-diastole frame, important for assessing Mitral Valve Stenosis (MS).
- **Teaching Assistant** Sep 2018 - Dec 2021
University of Science and Technology
 - * Served as a Python Programming Lecturer at various institutions. In this role, I taught foundational Python programming skills and introduced undergraduate and graduate students to a range of essential Machine Learning libraries, including NumPy, Pandas, and Matplotlib.
- **Machine Learning Engineer** Jul 2020 – Mar 2021
Gamelectronics
 - * Developed different DL models for the Data Management System (DMS).
 - * Utilized MySQL for effective database management, ensuring efficient data storage and retrieval within the DMS.

PROJECTS

- **Iris and Pupil Detection in Cataract Surgical Data**
 - * Description: Implemented YOLO object detection to identify iris and pupil in cataract surgery images for monitoring intraoperative deviations and optimizing post-operative outcomes.
 - * Tools & Technologies: YOLO Object Detection.
 - * Outcome: Achieved a Mean Average Precision (mAP) range of 0.75–0.80 across multiple datasets.
 - * GitHub Repository: YOLO Object Detection..
- **Classification of Surgeon Skill Level using V-JEPA and LSTM Models**
 - * Description: Classified surgeon skill levels based on surgical videos. The project utilizes the Cataract 101 dataset, leveraging both the V-JEPA model and an LSTM model with a ResNet-18 feature extractor.
 - * Tools & Technologies: LSTM and V-JEPA .
 - * Outcome: Achieved an accuracy of 83% and 93% using LSTM and V-JEPA models, respectively.
 - * GitHub Repository: V-JEPA: Video Joint Embedding Predictive Architecture.
- **Converting Handwritten Mathematical Equations to LaTeX Code**
 - * Description: Designed a system using image processing techniques and CNN models to classify symbols (digits, Latin/Greek letters, and mathematical operators) and convert them to corresponding LaTeX code.
 - * Tools & Technologies: CNNs.
 - * Outcome: Demonstrated potential of CNN models in detecting various mathematical symbols.
 - * GitHub Repository: Converting Handwritten Mathematical Equations to Latex Code.
- **Mid-distole Frame Detection using Echocardiography Images**
 - * Description: Applied image processing and machine learning algorithms to detect the mid-diastole frame in echocardiography images, aiding in mitral valve stenosis diagnosis.
 - * Tools & Technologies: MATLAB, K-Means, Circular Hough Transform .
 - * Outcome: Achieved a 1.40 average frame difference compared to the gold standard (echo-cardiologist opinion).
 - * GitHub Repository: Dimensionality Reduction (DR) methods.

SELECTED COURSES

- **Deep Learning for Computer Vision** 2024
University of Illinois at Chicago
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization,** 2023
Coursera
- **PyTorch for Deep Learning and Computer Vision** 2023
Udemy
- **Neural Networks and Deep Learning** 2020
Coursera

ACHIEVEMENTS & AWARDS

- **Award** 2023
Doctoral scholarship
Bioengineering Department
University of Illinois at Chicago, Chicago, IL
- **Award** 2015
Governmental scholarship
Department of Electrical Engineering
University of Science and Technology
- **Award** 2011
Governmental scholarship
Department of Electrical Engineering
Sahand University of Technology